PSPCL
PUNJAB STATE POWER
CORPORATION LTD.

Punjab State Power Corporation Limited (PSPCL)

Standard Operating Procedure

SOP 33: Cloud Security Management Procedure

Document Control

Document Title: SOP33 – Cloud Security Management Procedure

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Abbreviations

PSPCL	Punjab State Power Corporation Limited
IT	Information Technology
IaaS	Infrastructure as a Service
PaaS	Platform as a Service
SaaS	Software as a Service
CSP	Cloud Service Provider
MSP	Managed Service Provider
RBAC	Role Based Access Control
TLS	Transport Layer Security
VPN	Virtual Private Network
NGFW	Next Generation Firewall
NIPS	Network Intrusion Prevention System
HIPS	Host Intrusion Prevention System
WAF	Web Application Firewall
AD	Active Directory
DNS	Domain Name System
GoI	Government of India

1. Purpose

The purpose of this document is to provide framework for ensuring the confidentiality, integrity and availability of PSPCL assets such as data, applications and infrastructure hosted on cloud environment. It outlines the Data Classification, Data Encryption, Network Security Controls, Access Control, Incident Response and other measures that shall be implemented in the cloud environment for PSPCL data so that the compliance with relevant regulations is ensured and security risks are minimized.

2. Scope

This policy covers all the cloud services (viz. IaaS, PaaS, SaaS) being used by PSPCL IT department to host its Data, applications, services and infrastructure. It applies to all IT department personnel, CSPs, MSPs, firms and third-party vendors who access, manage, or use cloud-based resources on behalf of PSPCL.

3. Responsibilities

CSP, MSP, IT Department, End Users

4. Procedure

Proper security controls not limited to the following should be implemented on cloud environment to safeguard the PSPCL data or other related assets (such as applications, infrastructure) hosted by the CSP:

4.1. Access Control

4.1.1. The Access to cloud platform shall be assigned using Role Based Access Control (RBAC).

4.1.2. The access shall be provided after due approval from the concerned process owner.

4.1.3. The access shall be provided to users as per the principle of least privilege.

4.1.4. The user access shall be reviewed periodically.

4.1.5. The passwords shall be generated and provided to user as per PSPCL Password policy

4.2. Data Encryption

4.2.1. The sensitive data such as passwords, Aadhaar numbers shall be stored in the databases in encrypted form.

4.2.2. The data shall be encrypted while in transit using TLS protocol and VPN services.

4.2.3. The personal and sensitive data shall not be displayed on public platform without masking.

4.3. Network and Infrastructure Security Controls

4.3.1. Appropriate security solutions such as NGFW, WAF, NIPS, HIPS etc should be implemented to safeguard the data, applications and other assets hosted on cloud.

4.3.2. The policies and rules configured on security devices shall be reviewed periodically.

4.3.3. All network devices, servers, security devices, applications, databases hosted on cloud shall be patched with latest security patch.

4.3.4. Network segmentation shall be implemented for production and non-production environment.

4.3.5. The PSPCL data or application traffic shall be isolated and secured using DMZs, subnets, etc.

4.4. Incident Response and Plan

4.4.1. The CSP/MSP shall enable and maintain logs of all security soultions viz NGFW, NIPS, HIPS, End point security, WAF, system logs, AD, DNS, Application logs etc. The logs shall be retained for 180 days as per MoP guidelines.

4.4.2. The cloud environment shall be monitored and analyzed for detection of any threat or security incident.

4.4.3. The security incident management process shall be established by CSP/MSP for PSPCL cloud environment.

4.4.4. The occurrence of any security incident shall be reported to PSPCL within 6 hrs.

4.4.5. The CSP/MSP shall take appropriate action to ensure rapid response to the security incident.

4.4.6. The RCA (Root Cause Analysis) and the ATR (Action Taken Report) of the incident shall be submitted to PSPCL within 24 hours.

4.5. Compliance and Audits

4.5.1. The CSP/MSP shall ensure its compliance with GoI guidelines regarding cloud services and relevant ISO standards (including ISO 27001, ISO 27017 and ISO 27018).

4.5.2. The CSP/MSP shall provide support for internal as well as external audits conducted by PSPCL internal team or third party auditors.

4.5.3. The CSP/MSP shall conduct or provide support for periodic cyber security audits (including VAPT audit, configuration audits etc.). It shall also provide support for other requirements like forensic investigations as and when required.

4.5.4. The CSP/MSP shall ensure the closure of the vulnerabilities and compliance of advisories shared by PSPCL as reported by various government agencies CERT-In, NCIIPC, CEA, etc or third party auditors.

4.6. Enforcement

4.6.1. This policy shall comply with all other relevant IT policies of PSPCL.

4.6.2. Any employee from PSPCL who is found to have violated this policy may be subject to disciplinary action as per Employee Punishment & Appeal regulations of PSPCL.

4.6.3. If the CSP/MSP is found to be in violation of this policy, it may be subject to relevant IT laws and regulations, or face penalties in accordance with the terms outlined in its Work Order.

5. Document Review

5.1. The document shall be reviewed after every year or as when required with the prior approval as per prevalent instructions of PSPCL.

6. Reference

6.1. PSPCL ISMS Policy PO 33 for Cloud Security Policy